

Leaderboards Are Not All You Need: A Causal Check-Up for Driving Policies

Anonymous Submission
Paper ID —

Abstract—A policy’s rank on nuScenes’ open-loop metrics or NAVSIM’s leaderboard score predicts its rank at a new deployment site only about as well as a coin flip ($\rho = -0.54$ – $+0.49$ on near-pedestrian scenes): a benchmark score reports an outcome, not the behaviour behind it. When a vulnerable road user calls for a reaction, the score does not say whether that road user had any part in the outcome, so a ranking cannot separate a policy that yields from one that merely never gets close. We propose a check-up — a few hundred verified, pixel-level counterfactual edits (removing the pedestrian, re-rendering the scene at night) read as a diagnosis rather than a ranking. The edits reduce each scene to two axes: the displacement the pedestrian causes, and the displacement an irrelevant edit causes. Raw magnitude is then a poor safety signal. Replaying 246 right-hand-drive scenes at input speeds of 2–8 m/s, so that the pedestrian does lie on the planned path, only 2.5% of the responses that are called for qualify as genuine avoidance on a three-part test; on those cells neither the clearance to the pedestrian nor the planned distance moves. We pre-register a diagnosis made on 88 left-hand-drive frames and test it after the switch to right-hand drive: orderings transfer while point values do not, and the exam measuring how steadily a policy drives predicts the new side’s near-miss rate at $\rho = -0.94$ — a target none of the standard evaluations predicts. A check-up therefore prices a deployment verdict in tens of frames rather than in a new benchmark.

I. Introduction

A deployment decision is, implicitly, a bet that a benchmark rank still holds at the new site. The bet fails for a reason that is mechanical rather than statistical: a score is an outcome measure, and an outcome can be right for the wrong reason. A policy that never approaches a pedestrian because it plans short trajectories scores as well as one that sees the pedestrian and yields, and the two behave very differently the first time the scene changes. Telling them apart means asking why the score came out as it did, usually from little data: a closed-loop simulator for the target domain, or a fleet test, is exactly what the decision is meant to save.

We propose a check-up: a short list of exams, each with a control and a known normal range, whose results are read as a diagnosis rather than a ranking (Table I). Every exam is a counterfactual edit of real frames. We remove the pedestrian from the ego corridor, or re-render the frame at night, and read the change in the planned trajectory; an independent detector verifies every edit (Fig. 1). The central exam is a hazard-sensitivity curve, measuring how much farther the whole planned trajectory stays from the pedestrian’s logged future be-

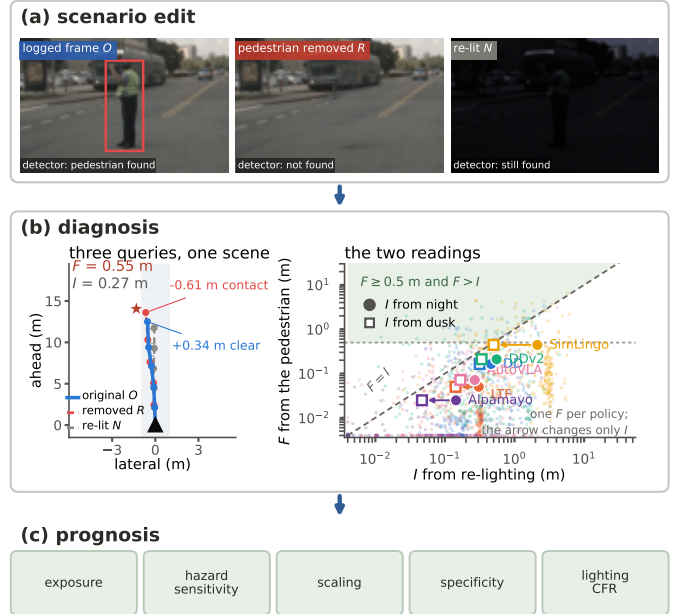


Fig. 1. The check-up, on one worked example. (a) A logged frame *O* and its two edits, pedestrian removed *R* and re-lit *N*, each checked by a detector no policy shares. (b) The policy is queried on all three; joining the plans at equal times gives two numbers, the displacement F the pedestrian causes and the displacement I the lighting causes, and the right panel places the policies on those axes. (c) These and the remaining readouts form the five exams of Table I.

cause the policy can see the pedestrian, against a model-independent hazard level. A specificity term applies a minimal-change principle: when no reaction is needed, the plan should not move.

a) Displacement is not avoidance: What an edit yields most directly—how far the plan moves when the pedestrian is removed—is a poor safety signal on its own, and treating it as one inverts the ranking. We decompose it into the magnitude of the response, the displacement the same policy produces under an edit unrelated to the pedestrian (its own jitter floor), and the separation the response actually buys. On 246 NAVSIM Singapore near-pedestrian scenes, the pedestrian outweighs the irrelevant edit in only 4–23% of scenes; among the cells where the plan did move, 43 moved away from the pedestrian and 42 moved towards it. The asymmetry that survives is one-directional and useful: a near-zero response certifies that the pedestrian did not enter the decision (93–99% of

scenes), while a large response certifies nothing.

b) The diagnosis: Applied to six end-to-end policies, the check-up separates quantities a benchmark score merges (Table II). Exposure varies $0.50\text{--}2.19\times$ across policies, and in five of six the safety benefit of seeing the pedestrian does not grow with the hazard, so collisions follow how far a policy plans to drive rather than what it perceives. A night rendering moves every policy more than the pedestrian does, along the interference direction $\hat{v}_I$, which opposes the faithfulness direction $\hat{v}_F$ governing the pedestrian response (Section VIII); Section IV traces the cause inside the networks.

c) The prognosis: A check-up is only useful if it generalises: it must be cheap, and what it finds in one domain must still hold in another. We price each verdict against sample size — most settle within a few dozen frames, hazard sensitivity within hundreds — then test generality directly, pre-registering seven predictions from a diagnosis on 88 Boston frames and checking them after the switch to right-hand drive. Orderings transfer ($\rho = 0.90$ against a pre-registered bar of 0.6); point values do not (Table VI), so we report rankings and verdicts rather than calibrated numbers. Re-measured on the new side, the exam that predicts behaviour there is not the one measuring how strongly a policy reacts but how steadily it drives ($\rho = -0.94$ for specificity), beating the leaderboard score computed on that side’s own scenes (Section V). The bet named above is therefore answerable: not by trusting a leaderboard rank, but by running a check-up whose predictive exams generalise and are known in advance.

Contributions:

- 1) A small-sample check-up for driving policies, built on verified, pixel-level counterfactual edits of real frames rather than the promptable text prompts language-model interpretability usually relies on — to our knowledge among the first uses of concept-vector-style methods on a multi-modal, largely non-promptable input. Its central result is methodological: raw response magnitude is a poor safety signal, and decomposing it into magnitude, jitter and separation turns a two-sided quantity into a one-sided certificate. We use it to diagnose six policies, with an explanation from inside the networks.
- 2) Evidence, in a new domain, for the linear representation hypothesis: the faithfulness direction $\hat{v}_F$ (response to the pedestrian) and the interference direction $\hat{v}_I$ (response to a change with no driving-relevant information) are geometrically opposed in aggregate (mean cosine -0.54 across four policies), and this opposition is mirrored behaviourally, not just in the representation. The relationship is a property of the pooled data, not a stable per-policy trait (Section VIII), and we report it as such.
- 3) An account of what small data can and cannot establish: sample-size laws for every exam and a pre-

TABLE I

The check-up. Each exam, the question it asks, the counterfactual that answers it, and the normal range of a driver that yields.

Exam	Question	Measurement	Normal
Exposure	How far ahead the plan commits	planned / logged distance, 2.5 s	≈ 1
Hazard sensitivity	Does it react to danger	safety gained by seeing the pedestrian, when needed	rises to 1 with hazard
HS	Does caution scale with danger	rank corr. of HS with hazard	> 0
Specificity	Does it over-react	plan change when no reaction is needed	≈ 1
SP	Does it react to noise	night vs. removal, same frame	CFR $\gg 1$, align ≈ 0
Lighting			
CFR, align			

registered left- to right-hand-drive transfer test. The exam that transfers predicts behaviour on the new side better than that side’s own leaderboard score does.

II. Related Work

a) Driving policies and their benchmarks: We examine BEV planners of the TransFuser lineage [1] (LTF, DiffusionDrive [2], DiffusionDriveV2 [3]) and vision-language-action models (SimLingo [4], AutoVLA [5], Alpamayo-1.5 [6], [7]). NAVSIM [8], [9] scores their plans by multiplying hard gates with a weighted average of soft terms. Open-loop scores are known to reward proxies [10]. We show that the six-policy spread is carried by a single gate.

b) Behavioural testing: Checklists of targeted behavioural tests [11] and corruption benchmarks [12]–[14] probe models beyond aggregate accuracy. A metric drop, however, does not say whether a policy became more cautious or more reckless. Our exams read signed plan changes against a counterfactual and state how many frames each needs.

c) Causal confusion: Imitation learners can respond to correlates of the expert’s action rather than to its causes [15]–[18]. Our exams apply this idea to trained policies: a cause (a pedestrian) and a nuisance (lighting) are intervened on directly.

d) Counterfactual editing and internal representations: Counterfactual explanations edit the scene [19], [20]; we verify every edit with a detector. Steering directions [21] can show that a network encodes what it does not act on [22]; we use them to explain a diagnosis, not as the diagnosis.

III. The Check-Up

A. Units, interventions and plans

Write a policy as a map π from a sensory-and-state input to a planned trajectory $X = (X_t)_{t \in [0, 2.5]}$, $X_t \in \mathbb{R}^2$ in the ego frame, sampled at 0.1 s. A unit is a pair

$u = (f, v)$: a logged frame f and an input ego speed v . We take v from the log for every frame and, wherever the pedestrian’s logged future is available, additionally $v \in \{4, 8\}$ m/s, which places the pedestrian on the planned path in scenes where the logged ego was stationary. The same grid is used on both corpora so that all six policies are read on one unit pool; the readouts are insensitive to it (widening the grid to $\{2, 4, 6, 8\}$ leaves the specificity and scaling orderings identical, $\rho = 1.00$).

Three interventions act on the input, leaving the policy untouched:

$$\mathcal{E}_O = \text{id}, \quad \mathcal{E}_R : \text{delete the pedestrian}, \quad \mathcal{E}_N : \text{re-light}, \quad (1)$$

and yield the paired plans $X^O = \pi(\mathcal{E}_O u)$, $X^R = \pi(\mathcal{E}_R u)$, $X^N = \pi(\mathcal{E}_N u)$. All quantities below are functionals of this triple and of the pedestrian’s logged future F_t . The horizon 2.5 s is the one all six policies share.

The displacement between two plans is

$$\bar{D}(X, Y) = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \|X_t - Y_t\|, \quad \mathcal{T} = \{0, 0.1, \dots, 2.5\}, \quad (2)$$

from which $F = \bar{D}(X^O, X^R)$ and $I = \bar{D}(X^O, X^N)$ follow: the plan change attributable to the pedestrian and the change the same policy produces under an edit that carries no driving-relevant information.

a) Realising $\mathcal{E}_R$: The target is localised by projecting its annotated 3D box into the camera rather than by taking the dataset’s 2D boxes, whose association to the tracked object is unreliable for the short-range, partially occluded pedestrians of these scenes. It is segmented with SAM [23] (box and torso prompts, clipped to the box, dilated by 9px) and inpainted with LaMa [24]; for the LiDAR policy the points inside the box are deleted. VLA policies read several cameras and only the front one is edited, so we retain only frames in which the target appears in no other view.

b) Realising $\mathcal{E}_N$: Each image $\mathbf{I} \in [0, 1]^{H \times W \times 3}$ the policy consumes is mapped by a closed-form photometric transform

$$\mathcal{E}_N(\mathbf{I}) = \text{clip}[g(\text{sat}_s \mathbf{I})^\gamma \odot \boldsymbol{\tau} + \sigma \boldsymbol{\epsilon}], \quad \boldsymbol{\epsilon} \sim \mathcal{N}(0, 1), \quad (3)$$

with sat_s the saturation scaling, $\boldsymbol{\tau}$ a per-channel tint and the noise seed fixed per frame. Two levels are used: night $(\gamma, g, s, \sigma) = (2.2, 0.45, 0.45, 7/255)$ with a blue tint, and the milder dusk $(1.45, 0.70, 0.80, 2/255)$ with a warm tint, the latter chosen to keep the pedestrian visible (Section VIII).

c) Validity of the interventions: An independent detector (Faster R-CNN [25], COCO, score ≥ 0.5) is run on the 248 front-only frames. It recovers the target in 100% of the original images, in 8.5% under $\mathcal{E}_R$ and in 89.9% under $\mathcal{E}_N$ (IoU ≥ 0.5): $\mathcal{E}_R$ suppresses the pedestrian for an observer outside the policy, while $\mathcal{E}_N$ leaves it visible and therefore changes the lighting rather than the scene. Two controls bound the residual artefact: pasting inpainting patches on empty road puts it at 5–17% of

the removal effect at pedestrian scale, and removing off-road pedestrians, which should not matter causally, gives a same-type null.

B. Readouts

With $c_t(X) = \|X_t - F_t\| - r_e - r_p$ the clearance net of the ego half-width $r_e = 1.0$ m and the pedestrian radius $r_p = 0.4$ m, a plan is summarised by four functionals,

$$S(X) = \text{mean}_t \min(c_t, \bar{c}), \quad A(X) = \mathbb{K}[\min_t c_t > 0], \quad (4)$$

$$C(X) = \text{clip}(\min_t c_t / d_0, 0, 1), \quad T(X) = \frac{1}{2.5} \inf\{t: c_t < d_0\}, \quad (5)$$

with $T = 1$ if the set is empty. The constants are physical, not tuned: r_e is the half-width of the nuScenes ego box (4.08×1.85 m), r_p the disc a standing adult occupies, $d_0 = 1$ m the outer edge of personal space [26], and $\bar{c} = 10$ m caps the contribution of scenes in which the pedestrian is far away. Equations (4)–(5) instantiate the standard surrogate-safety measures—contact, minimum clearance and an encroachment-time variant [27], [28]—read as distance to a person over the whole trajectory, as is conventional for navigation near people [29]. Speed is deliberately not a readout: the same deceleration is safe or unsafe depending on where the person is.

C. Exams

Let $L(X) = \sum_t \|X_{t+\delta} - X_t\|$ denote arc length and H the logged human trajectory.

a) Exposure: $\text{Exp} = \text{med } L(X^O) / L(H)$ over frames with $L(H) \geq 0.5$ m.

b) Hazard sensitivity: A unit needs a reaction when the pedestrian-blind plan enters personal space, $C(X^R) < 1$. On those units, writing $\Delta(\cdot) = (\cdot)(X^O) - (\cdot)(X^R)$,

$$\text{HS} = \frac{1}{4} [\Delta A + \Delta C + \Delta T + \tanh(\Delta S / d_0)] \in [-1, 1]. \quad (6)$$

Each term lies in $[-1, 1]$ by construction and $\tanh$ squashes the single unbounded one at the d_0 scale. The composite is ours; its ingredients are not. Its four terms are the surrogate-safety measures of (4)–(5), and averaging normalised sub-scores into one index is how the benchmark we compare against forms EPDMS [8], keeping the two commensurable. What the check-up adds is that every sub-score is a paired difference across $\mathcal{E}_R$ —the ablation form used to attribute behaviour to an input in interpretability work [21], [30]. HS is read against the model-independent hazard

$$a_{\text{req}} = \frac{v^2}{2 \max(d - 2, 0.5)}, \quad \text{TTC}_0 = \frac{d - 2}{v}, \quad (7)$$

the deceleration to stop short of a pedestrian at distance d and its time-based counterpart [27], [28]. Units with $v < 1$ m/s are excluded: a stationary ego can move neither toward nor away from anyone.

c) Scaling and specificity: $Sc = \rho(HS, a_{req})$ over needed units, with ρ Spearman’s rank correlation. On units needing no reaction, the minimal-change principle gives

$$SP = \exp(-\bar{D}(X^O, X^R)/0.5\text{ m}) \in (0, 1], \quad (8)$$

so $SP \rightarrow 1$ iff an irrelevant edit leaves the plan where it was.

d) Lighting:

$$CFR = \frac{\mathbb{E} \bar{D}(X^O, X^R)}{\mathbb{E} \bar{D}(X^O, X^N)} = \frac{\mathbb{E} F}{\mathbb{E} I}, \quad (9)$$

$$\text{align} = \text{med} \cos(X^N - X^O, X^R - X^O). \quad (10)$$

A policy for which lighting is irrelevant has $CFR \gg 1$ and $\text{align} \approx 0$; positive alignment means the re-lighting pushes the plan the way deleting the pedestrian does. In Section VIII the same two contrasts are read in representation space as unit directions $\hat{v}_F$ (faithfulness) and $\hat{v}_I$ (interference), and align is their cosine.

D. Policies and corpora

We examine LTF and DiffusionDrive (front camera and ego status), DiffusionDriveV2 (camera and LiDAR), SimLingo (InternVL2-1B), AutoVLA (Qwen2.5-VL-3B; three cameras $\times$ four frames) and Alpamayo-1.5 (four cameras; a diffusion head over the VLM cache), all with released weights. DiffusionDrive’s diffusion seed is fixed so that paired queries share their noise.

The primary corpus is nuScenes [31] keyframes: corridor pedestrians (lateral clearance to the ego corridor $\leq 1\text{ m}$) at 5–12m, their off-road counterparts, and 30 approach sequences at 5.8–14.6m with logged pedestrian futures. The near-pedestrian units come from 236 front-only frames, 88 in Boston (left-hand drive) and 148 in Singapore (right-hand drive); Alpamayo-1.5 covers 170 of the 236, needing 1.6s of history and 6.4s of future within the scene. Section VII repeats $\mathcal{E}_R$ and $\mathcal{E}_N$ on a second, right-hand-drive corpus—the NAVSIM test split [8], 254 Singapore scenes with a pedestrian near the ego corridor, of which 246 carry a complete logged pedestrian future—used only for the transfer and decomposition results; Table II is measured on the nuScenes frames alone.

IV. Diagnosis of Six Policies

A. The report

Table II and Fig. 3b give the six profiles.

a) Exposure: The six policies do not even agree on how far to plan, and this alone will turn out to explain more of the collision picture below than perception does: exposure spans a factor of four, from DiffusionDrive at half the human’s distance to SimLingo at more than twice it, with AutoVLA and Alpamayo-1.5 near the human’s own.

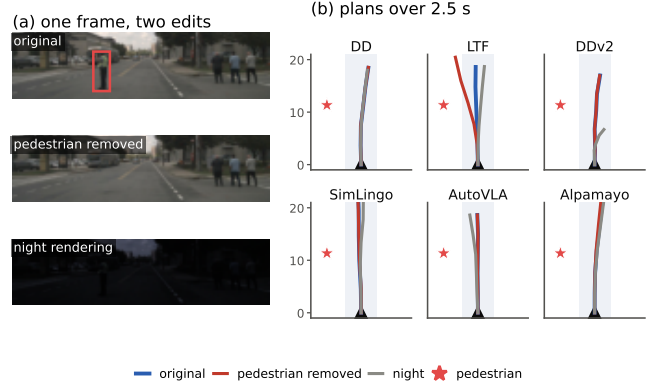


Fig. 2. One frame, two edits, six policies. (a) A worker in a high-visibility vest at the edge of the ego corridor, 9.6 m ahead, ego speed 7.6 m/s; the detector finds the worker in the night image but not in the removed one. (b) Plans over 2.5 s under each input: original (blue), pedestrian removed (red), night (grey); the star marks the worker.

TABLE II

The check-up report. Normal range in brackets (Table I); $\downarrow/\uparrow$ = outside it, bold = best in column. Collision: contact with the pedestrian’s logged future at 8 m/s input speed, visible / removed.

Policy	exposure [0.8, 1.2]	HS [≥ 0.15]	scaling [> 0]	SP [≥ 0.8]	CFR [> 1]	coll. 8 m/s vis. / rm.
DiffusionDrive	0.50 $\downarrow$	0.06 $\downarrow$	-0.28 $\downarrow$	0.76 $\downarrow$	0.23 $\downarrow$	7.6 / 9.2
LTF	0.79 $\downarrow$	0.03 $\downarrow$	-0.13 $\downarrow$	0.78 $\downarrow$	0.23 $\downarrow$	9.9 / 10.7
DiffusionDriveV2	0.88	0.10 $\downarrow$	+0.23	0.65 $\downarrow$	0.47 $\downarrow$	3.1 / 6.1
SimLingo	2.19 $\uparrow$	0.00 $\downarrow$	-0.22 $\downarrow$	0.44 $\downarrow$	0.23 $\downarrow$	40.5 / 36.6
AutoVLA	1.01	0.02 $\downarrow$	-0.18 $\downarrow$	0.76 $\downarrow$	0.44 $\downarrow$	32.1 / 29.8
Alpamayo 1.5	1.13	0.01 $\downarrow$	-0.23 $\downarrow$	0.84	0.25 $\downarrow$	11.2 / 11.2

b) Hazard sensitivity: No policy manages the pedestrian risk. On units that need a reaction, seeing the pedestrian buys 0.00–0.10 of safety on a $[-1, 1]$ scale. The benefit does not grow with the hazard: its rank correlation with a_{req} is negative for 5 policies, down to -0.28 for DiffusionDrive; DiffusionDriveV2 is the only exception ($+0.23$), and its benefit stays below 0.15. Replacing the hazard axis by the time-based TTC_0 , whose sign is reversed by construction, leaves the picture unchanged (-0.07 to $+0.32$, against -0.28 to $+0.23$). The benefit is also unreliable in direction: counting only cases where seeing the pedestrian changes the outcome, it turns a collision into a clean pass about as often as it turns a pass into a collision, for every policy (DiffusionDrive 9 vs. 5, SimLingo 18 vs. 30, the rest at most 5 either way; all McNemar $p \geq 0.11$) — what reads as a safety margin in Table II is, case by case, close to a coin flip.

c) Specificity: A policy that moves when nothing requires it is not merely cautious, it is unpredictable, and unpredictability is what turns out to travel across a domain shift (Section VI). SimLingo’s plan moves most when nothing requires it ($SP = 0.44$); the other five stay steadier, at 0.65–0.84.

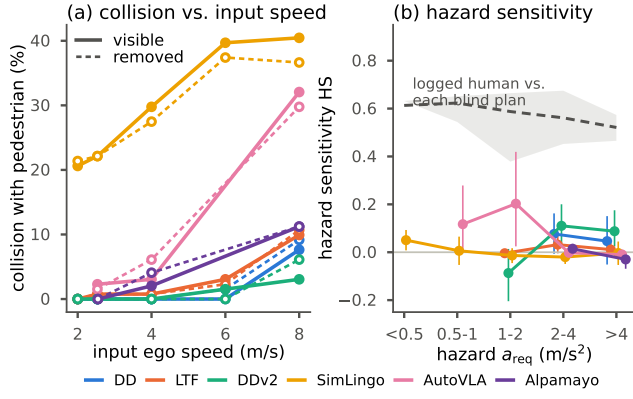


Fig. 3. Collision and hazard sensitivity. (a) Contact between the plan and the pedestrian’s logged future, at the logged speed and at counterfactual input speeds; solid: pedestrian visible, dashed: removed. (b) HS against the hazard level a_{req} (mean, 95% CI, bins with ≥ 8 units; points dodged within a bin, and a policy appears only where it has enough units). Dashed: the logged human trajectory scored against each policy’s own blind plan, averaged over policies; band: the range across the six.

TABLE III

Inside the BEV planners. Top: the policy’s own BEV semantic head on 520 NAVSIM near-pedestrian scenes (in parentheses: the same readout at a mirrored empty location). Middle: difference-of-differences injection (corridor minus off-road pedestrian direction), reported as effect over the median of random directions, with permutation p . Bottom: planning reach.

	DD	LTF	DDv2
Ped. peak at pedestrian / global	0.07 (0.11)	0.07 (0.07)	0.09 (0.15)
Ped.-box px won by ped.	0%	0%	0%
Pedestrian / vehicle peak	0.018	0.023	0.052
Injection, 8m grid (plan-ner)	$2.9 \times (p=.27)$	$p=1.0$	–
Injection, 1m grid (branch)	$52 \times (p=.007) \equiv 0 (14/14)$	–	–
effect vs. 8m grid	$\sim 1/460$	–	–
Median plan length (m)	5.3	7.1	8.9
Median pedestrian distance (m)	20.8	20.8	20.6
Pedestrian within plan reach	5%	11%	10%

d) Collisions: Collisions with the pedestrian’s logged future follow how far a policy plans to drive, not what it perceives. At the logged speed, the visible and removed rates differ by at most 0.4 points for every policy, and at 8 m/s by at most 3.8. SimLingo, which plans the longest trajectories, collides most. The low-exposure policies almost never reach the pedestrian. DiffusionDrive is safe because it stays short, AutoVLA and Alpamayo-1.5 cover a human-like distance without yielding, and SimLingo covers the most ground with the least stable plan.

B. Looking inside: why the risk goes unmanaged

The check-up only diagnoses; the following explains the diagnosis for the three BEV planners (Table III, Fig. 3).

a) The pedestrian leaves a trace but is not decoded: The planners’ own BEV semantic heads do not localise the pedestrian. On 520 NAVSIM near-pedestrian scenes, the pedestrian channel is no brighter at the pedestrian than at a mirrored empty location (Table III, top). It is 20–55 times weaker than the vehicle channel and never wins the argmax inside a pedestrian’s box. The class is not merely inert: DiffusionDriveV2 does assign pixels to it in 3 of the 520 scenes, but those pixels sit about 5 m from the pedestrian, and DiffusionDrive and LTF never assign it a pixel anywhere. The pedestrian does leave a trace in the features, however. The direction between SimLingo’s representations with and without the pedestrian is steerable: it shortens the plan by 2.9 m at the largest dose, while random directions lengthen it. Location-specific information also survives in the 1 m side branch, as shown next.

b) The planner cannot locate it: The trajectory head reads an 8×8 grid at 8 m per cell. In 73% of 94 pairs of corridor and off-road pedestrians, the two fall in the same or an adjacent cell. A corridor-minus-off-road direction injected there does not beat random directions ($p = 0.27$). At the 1 m side branch it does (52-fold, $p = 0.007$), but with $\sim 1/460$ of the effect on the trajectory; in LTF that branch does not reach the trajectory at all.

c) The plan does not reach the pedestrian: Plans end 5–9 m ahead, while the median corridor pedestrian stands 20 m away; only 5–11% of pedestrians are within reach. A distance sweep over 850 nuScenes scenes (974 pedestrian records) confirms that the limit is geometric rather than perceptual: once the pedestrian is within 12 m, LTF begins to respond, and at 8–12 m its plan changes by 0.23 m ($p = 5 \times 10^{-7}$), which separates corridor from off-road pedestrians after mask area is regressed out (AUC 0.65; 0.67 on an independent draw). Raising the input speed removes that response again, and collisions then rise identically with and without the pedestrian in view (Fig. 3a). DiffusionDrive never responds at any distance.

V. Beyond the Leaderboard: What the Check-Up Predicts

These policies are usually compared by a leaderboard rank or an open-loop error. Running those scores and the check-up on the same six policies (Table IV) lets us ask what each recovers that the other does not, and which of them predicts behaviour at a site it has not seen.

a) nuScenes open-loop metrics: On our 236 near-pedestrian frames, removing the pedestrian from the image changes none of the standard open-loop numbers: the L2 error by at most 0.07 m, the frontal collision rate by at most 1.7 points and the pedestrian collision rate by at most 3.0 (Table IV shows the L2 pairs; the two collision rates behave the same way and are omitted). The metric cannot

TABLE IV

The same six policies under the standard scores. Left: nuScenes open-loop L2 on the 236 near-pedestrian frames, original / pedestrian removed. Middle: NAVSIM EPDMS on all 783 Singapore scenes and on the 260 with a pedestrian near the corridor. Right: share of moving-ego scenes with minimum pedestrian TTC < 1.5s, by driving side, with the rank in parentheses. Bold: best in column.

Policy	L2 (m)	EPDMS		TTC<1.5s	
	orig. / rm.	all	near	LHD	RHD
DD	0.98 / 0.99	0.535 (4)	0.744 (2)	3.2 (1)	14.1 (1)
LTF	0.88 / 0.91	0.509 (5)	0.735 (3)	5.4 (2)	16.5 (2)
DDv2	1.35 / 1.31	0.652 (3)	0.367 (6)	18.3 (3)	22.4 (5)
SimLingo	3.28 / 3.35	0.359 (6)	0.583 (4)	61.3 (6)	56.5 (6)
AutoVLA	0.68 / 0.66	0.683 (1)	0.413 (5)	19.4 (4)	21.2 (4)
Alpamayo 1.5	0.77 / 0.72	0.663 (2)	0.760 (1)	25.4 (5)	20.0 (3)

tell whether a policy saw the pedestrian. Its ordering follows how far a policy’s planned distance departs from the human’s (rank correlation +0.66 with $|\text{exposure} - 1|$): L2 rewards driving at the human’s speed. We exclude rear-end contacts from the collision count. The logged traffic does not react, so a policy that plans to drive slower than the human is scored as being hit from behind.

b) NAVSIM, sliced to near pedestrians: On all 783 Singapore scenes, AutoVLA, Alpamayo-1.5 and DiffusionDriveV2 score highest. On the 260 scenes with a pedestrian near the ego corridor, the ordering is unrelated to that one ($\rho = -0.09$). The two rank columns of Table IV can be read side by side: the leaderboard’s first (AutoVLA) falls to fifth and its third (DiffusionDriveV2) to last, while its fourth and fifth (DiffusionDrive, LTF) rise to second and third. Whichever policy a leaderboard reader would have picked, the near-pedestrian scenes would have picked another. In these scenes the no-collision gate carries 100% of the spread, while over all 783 scenes it carries 1% and the two map-compliance gates carry 95% (drivable area 76%, driving direction 19%): the leaderboard ranks these policies by map compliance, and none of the exams above is visible in it. The cause is a trait the check-up had not examined: 63% of these scenes show a human driver waiting. DiffusionDriveV2 and AutoVLA plan to move off and run into the stopped traffic. On our own frames where the human waits, DiffusionDriveV2 plans to move off in 81% of cases, against 3% for DiffusionDrive and 0% for Alpamayo-1.5. This agrees with NAVSIM for the extremes but not for AutoVLA. Of our exams, specificity predicts the near-pedestrian ordering ($\rho = +0.66$), while hazard sensitivity does not (+0.14), because no policy has any.

c) Pedestrian TTC across driving sides: Following the plan against the pedestrian’s logged future, we count the moving-ego scenes in which the minimum time to collision falls below 1.5s. Near-pedestrian scenes from the left-hand-drive cities (nuScenes Boston and NAVSIM Las Vegas, Boston, Pittsburgh) and from Singapore give orderings that agree at $\rho = +0.77$. Both orderings coincide with planned distance ($\rho = +1.00$ and +0.77 with exposure).

TABLE V

Which evaluation predicts behaviour near pedestrians in another domain? Rank correlation between the six-policy ordering given by each evaluation (none uses the target data) and the ordering observed on 260 NAVSIM Singapore scenes with a pedestrian near the ego corridor (original images, moving ego). Bold: $|\rho| \geq 0.89$ (two-sided 5% for $n=6$).

Ordering from	Singapore near-pedestrian ordering by			
	TTC	clearance	collisions	EPDMS
Same readouts, nuScenes Boston				
TTC violations	+0.94	+0.83	+0.94	+0.14
clearance	+0.83	+0.94	+0.83	+0.60
collisions	+0.83	+0.94	+1.00	+0.09
Same readouts, NAVSIM US cities				
TTC violations	+0.83	+0.94	+1.00	+0.09
clearance	+0.83	+0.71	+0.89	-0.03
near-ped. EPDMS	+0.77	+0.54	+0.43	+0.89
nuScenes open-loop (Boston)				
L2	+0.43	+0.09	+0.03	+0.49
collision	+0.26	-0.09	-0.03	+0.03
ped. collision	+0.60	+0.37	+0.26	+0.71
NAVSIM leaderboard				
EPDMS, 783 scenes	+0.14	-0.09	-0.03	-0.09
Check-up (Boston)				
exposure	+0.83	+0.94	+1.00	+0.09
hazard sensitivity	-0.66	-0.89	-0.94	+0.09
specificity	+0.83	+0.71	+0.60	+0.77
lighting CFR	-0.20	-0.09	+0.14	-0.94
patience	+0.31	-0.03	-0.14	+0.77

They agree only partly with hazard sensitivity (+0.83, +0.49), which is near zero for every policy, and not with lighting invariance (+0.14, -0.43). NAVSIM’s own TTC term, which counts all objects over 4s, agrees only weakly (0.31–0.54).

d) Which evaluation predicts behaviour in another domain?: This is the question a deployment decision has to answer, so we pose it directly in Table V. The target is the ordering of the six policies on the Singapore near-pedestrian scenes (original images), by TTC violations, clearance to the pedestrian, pedestrian collisions and EPDMS. Each predictor is an ordering obtained without that data. The standard evaluations do not predict it: nuScenes L2 and collision rates, and the NAVSIM leaderboard score, correlate with it at -0.54 to +0.49. The same pedestrian-centred readouts taken on Boston or on the US cities do predict it (0.83–1.00), and so does the check-up’s exposure exam (0.94). Near-pedestrian EPDMS is predicted best by near-pedestrian EPDMS elsewhere (0.89) and by specificity (0.77). Hazard sensitivity and lighting invariance measure something else, and they do not predict these orderings.

e) Reading: Where a standard metric measures what an exam measures, they agree (pedestrian collisions: $\rho = +0.94$; TTC and exposure). None shows whether the pedestrian changed the plan, and the full-set score hides the near-pedestrian scenes.

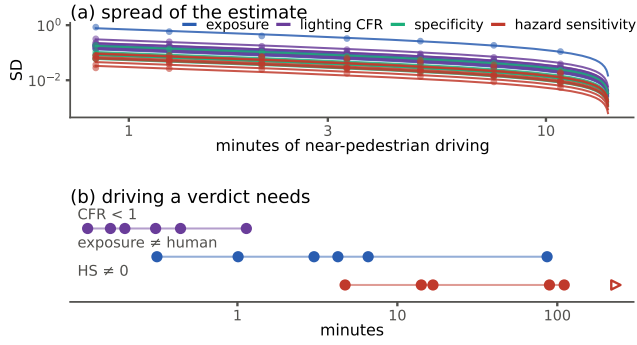


Fig. 4. How much driving a verdict costs. One frame is 5 s of near-pedestrian driving. (a) Spread of each exam’s estimate against sample size, with its fitted law; one line per policy. (b) Driving needed to settle each verdict, one dot per policy; values beyond the axis are off scale.

VI. How Much Data Does a Check-Up Need?

A. Fitting

For every exam and policy we subsample n of the 170 frames that all six policies share, 400 times, and fit the spread of the estimate with $\sigma(n) = a\sqrt{1/n - 1/N}$ (Fig. 4).

a) Well-fitting exams: Exposure, the lighting ratio and specificity obey this law ($R^2 = 0.97\text{--}1.00$). The fitted a converts directly into the number of frames a verdict needs.

- Lighting outweighs the pedestrian ($\text{CFR} < 1$) is settled for every policy within 14 frames.
- Exposure differs from the human’s is settled within 51 frames for the four policies whose exposure is far from one. For AutoVLA and Alpamayo-1.5 it would take hundreds to thousands of frames; this is itself a finding, since they plan a human-like distance.
- Orderings of the six policies stabilise quickly: exposure with 10 frames (96% of subsamples reach $\rho \geq 0.6$ with the full ordering) and specificity with 40 (99%).

b) Poorly fitting exams: Hazard sensitivity and the direction of the lighting effect behave differently. Their fits are poor (R^2 as low as 0.26 and -0.09) wherever few units need a reaction. Deciding that $\text{HS} \neq 0$ needs 57–1,326 frames, except for SimLingo, whose estimate sits on the threshold, and the ordering reaches 84% stability only at 90 frames. The ordering by alignment is unstable even at 130 (72%).

In wall-clock terms, a 20 s scene in this corpus yields about 4 usable near-pedestrian frames, so the lighting verdict costs roughly 3 scenes and the exposure verdict 13; deciding that hazard sensitivity differs from zero costs 331—about 110 minutes of logged driving that actually contains a pedestrian near the corridor.

The division is therefore clear. A few dozen frames diagnose planned distance, over-reaction and sensitivity to lighting. Whether a policy manages hazard, and in which

TABLE VI

Pre-registered transfer test. Predictions fixed on 88 Boston frames before any Singapore statistic was computed, then tested on 148 Singapore frames. Five policies; keeping the replaced sixth changes no verdict (Section VI).

Prediction	Holds	Evidence
P1 Lighting outweighs the pedestrian ($\text{CFR} < 1$) for every policy	yes	max upper CI 0.70
P3 Exposure ordering transfers ($\rho \geq 0.6$; SimLingo top, DD bottom)	yes	$\rho = 0.90$
P4 Over-reaction ordering transfers ($\rho \geq 0.6$; SimLingo lowest SP)	yes	$\rho = 0.60$
P7 SimLingo collides most; visible vs. removed within 3 pts	yes	gap 2.7 pts
P2 Hazard sensitivity stays below 0.15 (upper CI)	no	fails for DD, DDv2
P5 Point values stay within the left-hand CI ($\geq 70\%$ of 15 cells)	no	60%
P6 Purer policies (lower $ \text{align} $) shift less	no	$\rho = -0.90$ (opposite)

direction lighting pushes it, needs hundreds of frames that contain a genuine need to react.

B. A pre-registered transfer test

Does a diagnosis made on one driving side describe the same policy on the other? Before computing any right-hand-drive statistic, we fixed seven predictions and their decision rules. They were derived from the profile of the 88 Boston frames, and we tested them on the near-pedestrian units of the 148 Singapore frames (Table VI). The registration covered the six policies public at that time. NVIDIA subsequently released Alpamayo 1.5, which accepts navigation instructions, and we use it in place of Alpamayo-R1 throughout this paper; because both driving sides of Alpamayo 1.5 were measured together, its numbers do not satisfy the registration discipline, so the table is evaluated on the five policies that do. Retaining the replaced policy changes none of the seven verdicts.

a) What transferred: 4 of the seven predictions hold, and they are the orderings and verdicts. The night rendering outweighs the pedestrian in every policy. Exposure keeps its order ($\rho = 0.90$), and so does over-reaction, with SimLingo the most unstable. SimLingo collides most, with the visible and removed rates within 3 points of each other.

b) What did not: Three predictions fail.

- Point values drift: only 60% of the model-exam cells stay inside their Boston interval.
- In Singapore, DiffusionDrive and DiffusionDriveV2 show a small but non-zero hazard sensitivity ($\text{HS} = 0.09$ and 0.08).
- The “purity” hypothesis is reversed. We had predicted that policies whose lighting effect is more orthogonal to their pedestrian response would shift less between sides; the observed rank correlation is -0.90 . Alignment is itself unstable between the two sides (Section VIII).

TABLE VII

CFR and the two rules across driving sides. CFR: the check-up value on the nuScenes frames of Table II, Boston and Singapore separately, and the median over the ten cases taken per scene. TTC and the rule columns come instead from the 246 NAVSIM Singapore scenes all six policies share; rules are given forward / reversed.

Policy	CFR			TTC	rule holds %	
	Bos.	Sing.	scene	<1.5 s	P-F	P-I
DD	0.17	0.24	0.03	13%	95/74	95/97
LTF	0.34	0.19	0.12	16%	96/100	97/100
DDv2	0.39	0.52	0.13	22%	96/67	96/87
SimLingo	0.32	0.19	0.06	56%	93/73	95/91
AutoVLA	0.41	0.46	0.00	21%	95/60	98/86
Alpamayo 1.5	0.15	0.26	0.13	20%	99/33	94/93

c) Reading: A small check-up can therefore report an ordering and a verdict for use in a new domain, but not a calibrated value.

VII. Re-measuring F and I in the New Domain

Section VI moved a diagnosis from one driving side to the other. The more useful question for deployment is whether the method moves: can a handful of counterfactual cases in the new domain characterise how the policies behave there? We repeat both interventions on the 246 NAVSIM Singapore near-pedestrian scenes and read, per scene, the two behavioural displacements that correspond to $\hat{v}_F$ and $\hat{v}_I$ at the output: F , the plan displacement caused by removing the target pedestrian, and I , that caused by the night rendering.

a) What F and I say per scene: Across the ten detailed cases and six policies (60 cells), the median F is 0.05 m and the median I is 0.62 m; the lighting moves the plan more in 85% of cells. Two rules follow from the axes. We state each in both directions, so that every cell receives a prediction:

- P-F: if $F < 0.5$ m, removing the pedestrian should leave the clearance and the contact outcome unchanged; if $F \geq 0.5$ m, it should change them.
- P-I: if CFR < 1, the night rendering should move the outcome at least as much as the removal; if CFR ≥ 1 , the removal should move it at least as much as the night.

The two directions do not fare equally. On the detailed cases, P-F holds in 54/55 cells in the forward direction but only 2/5 in the reverse; P-I holds in 51/51 and 8/8. Scaling both interventions to the full pool, which all six policies cover, reproduces the asymmetry exactly (Table VII): P-F holds in 93–99% of scenes forward against 33–100% reversed, while P-I holds in 94–98% and 86–100%. A near-zero F is therefore a reliable certificate that the pedestrian is not being used, but a large F is not a promise of safer behaviour.

b) When the plan does move, which way does it move?: The reverse direction fails for an instructive reason: F measures how far a plan moves, not where it

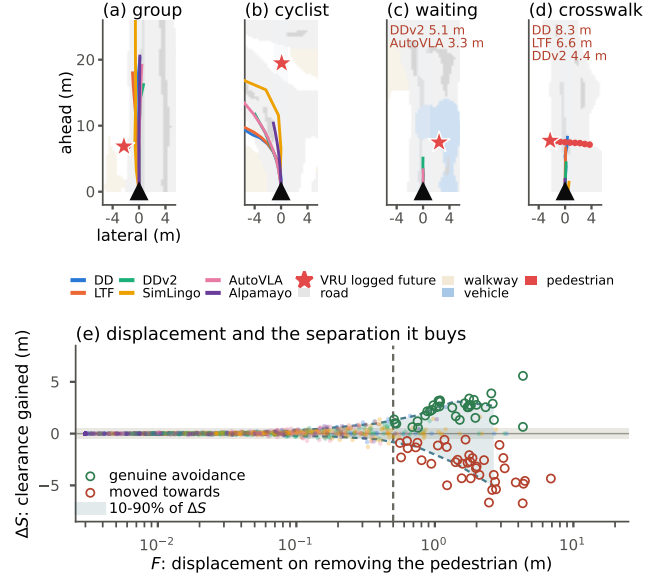


Fig. 5. Four right-hand-drive cases, and displacement against the separation it buys. (a–d) Ego at the origin, the VRU’s logged future dashed from the star, and DiffusionDriveV2’s own BEV semantic head as the background; every policy receives the frame’s logged navigation, so the turns are commanded. (a) Clearances 0.3–0.9 m. (b) SimLingo passes at 3.2 m, the other five at 6.7–10 m. (c, d) The departing policies and their planned distances are named in the panel; Boston departure rates are DiffusionDriveV2 81%, AutoVLA 9%, DiffusionDrive 2.8%. (e) Every policy–scene cell of the 246 Singapore scenes: F against ΔS ; band: the 10–90% spread of ΔS per F bin; dashed line: $F = 0.5$ m.

moves to. We therefore decompose it (Fig. 5e), reading F against I —the same policy’s jitter floor under an unrelated edit—and against ΔS , the change in clearance to the pedestrian’s logged future. A response counts as avoidance only if all three hold: the plan moved ($F \geq 0.5$ m), it moved more than the policy’s own jitter ($F > I$), and the movement bought distance ($\Delta S \geq 0.5$ m).

Three readings follow. First, the pedestrian outweighs the irrelevant edit in only 4–23% of scenes, so most of what F registers sits inside the policy’s own variability—the pinched band of Fig. 5e. Second, avoidance is rare where it is called for. The logged ego is stationary through most of this corpus, so a plan that does not move is usually the correct one; we therefore replay every scene at input speeds of 2–8 m/s, as in Section III, and read only the cells in which the pedestrian-blind plan does come within 1 m of the pedestrian’s logged future. Across 4 policies that is 446 of 2952 cells (11–24% per policy), and 11 of them—2.5%—are genuine avoidance. On those cells the median clearance changes by +0.00–+0.03 m and the median planned distance by −0.02–+0.08 m: facing a pedestrian it would otherwise reach, no policy diverts and none slows. Third, the displacement that does occur is not aimed: of the 121 cells in which the plan moved, 43 increased the clearance and 42 decreased it. Per policy the split differs

sharply: DiffusionDrive reacts in few scenes but in the right direction (median $\Delta S = +2.16$ m), while SimLingo moves in more scenes than the other five combined and moves towards the pedestrian more often than away (32 against 13, median -0.59 m); it also has the highest rate of sub-1.5 s time-to-proximity (Table VII). Finally, what a policy yields does not grow with the hazard: the rank correlation between ΔS and the deceleration the scene demands is -0.37 – $+0.07$.

An exam that rewarded responsiveness alone would therefore rank the least safe policy first.

c) What a small sample in the new domain buys: We draw k scenes at random from the pool, re-measure F and I on them and compare against the large-sample values, 200 draws per k . Ordering and level behave very differently. The ordering of how dangerously the policies behave—the share of scenes whose minimum time-to-proximity falls below 1.5 s—is recovered almost immediately: $\rho = +0.64$ at $k = 10$ and $\rho = +0.65$ at $k = 20$ ($\rho \geq 0.6$ in 70% of draws), rising to $\rho = +0.76$ at $k = 40$. The level is not: the same $k = 20$ sample misstates the violation rate by 13 points and CFR by 0.116; the rule hit-rates sit in between, at 3.6 points. Importing the Boston CFR values instead misstates the Singapore ones by 0.099, already worse than re-measuring $k = 40$ fresh scenes (0.078), while 80 scenes reach 0.054 and 160 0.027. Values measured elsewhere are worth about forty scenes measured here.

d) Which exam travels: Of the exams in the report, the one that predicts the new domain’s behaviour is not the one that measures how much a policy reacts, but the one that measures how steadily it drives. Correlating each Boston exam against the Singapore rate of sub-1.5 s time-to-proximity over the six policies, specificity gives $\rho = -0.94$ —the only value significant at the two-sided 5% level for $n=6$ —against $+0.77$ for exposure and $+0.37$ for CFR. Specificity asks how far a plan moves when an irrelevant part of the scene is edited; a policy that answers “not at all” drives steadily, and steadiness is what survives the change of side. (Exposure travels too: on the larger original-image set of Section V it reaches 0.94 against the same target. Specificity is the stronger predictor and the only one that clears significance here.) The decomposition above says why: displacement that is not aimed away from the pedestrian is not protection, and a policy producing a lot of it is dangerous on both sides.

e) What transfers and what must be re-measured: The verdict $\text{CFR} < 1$ holds on both sides for every policy, but the value drifts by a factor of 0.6 to 1.7 and its across-policy ordering barely agrees between sides ($\rho = +0.37$, against $+0.83$ for specificity); per-scene medians are weaker still (Table VII). The instrument transfers while its readings do not.

TABLE VIII

What the night rendering does to the plan. CFR: plan change from removing the pedestrian over that from the night rendering, same frames (95% CI; ideal $\gg 1$). Speed: change of planned mean speed under the night rendering. ΔS : change of whole-plan separation from the pedestrian (negative = closer). Align: direction of the night-induced change relative to the removal-induced one. Bold: $p < 0.01$.

Policy	CFR [95% CI]	speed	ΔS (m)	align
DD	0.30 [0.20,0.43]	+37%	−0.37	+0.55 → +0.09
LTF	0.36 [0.24,0.49]	+12%	−0.24	−0.09 → −0.42
DDv2	0.49 [0.33,0.70]	−8%	+0.15	+0.63 → +0.16
SimLingo	0.31 [0.21,0.44]	+16%	−0.47	+0.30 → +0.14
AutoVLA	0.38 [0.21,0.57]	−0.3%	+0.005	+0.65 → +0.62
Alpamayo 1.5	0.18 [0.10,0.30]	−2%	+0.03	+0.81 → −0.30

VIII. Lighting: Magnitude, Direction, and the Pedestrian

Lighting should not change what a driver does, so this exam asks how far a policy is moved by an irrelevant change. It is the most robust result in the check-up and the one whose direction is least stable.

a) Magnitude: Removing a pedestrian from the corridor moves every policy’s plan only 0.18–0.49 times as far as the night rendering does, with every 95% interval below one (Table VIII); on detector-verified edits $\text{CFR} = 0.30$ –0.64, and only Alpamayo-1.5’s interval (45 frames) reaches above one. On a separate pool of 312 lead-vehicle events the effect is carried by sensor noise rather than by darkness: for DiffusionDrive, darkening alone raises the planned speed by $+0.08$ m/s and darkening with noise by $+2.05$ m/s. It is also global—perturbing only the sky recovers 19% of the effect of deleting the lead vehicle.

b) Two lighting levels: The exam re-lights at two strengths, the milder one chosen to keep the pedestrian visible. At night the detector still finds 67% of the 45 targets it finds in the original, at dusk 89% (removal: 2%). On the same 246 scenes the ratio is 0.15–0.38 at night and 0.34–0.87 at dusk: below one for every policy at both levels, with the pedestrian in plain view.

c) Direction: faster, and therefore closer: A policy that slows at night is doing something defensible; one that speeds up is not. Three policies plan to go faster under the night rendering: DiffusionDrive by 37%, SimLingo by 16% and LTF by 12%, and their whole plans pass 0.24–0.47 m closer to the pedestrian. The approach is longitudinal: the further the plan’s endpoint moves forward, the closer it gets (rank correlation -0.73 to -0.96), while the endpoint shifts toward the pedestrian’s side no more often than chance. DiffusionDriveV2, the only policy that also reads LiDAR, slows (−8%) and keeps more distance; AutoVLA and Alpamayo-1.5 are unaffected.

d) The two directions in representation space: We write $\hat{v}_F$ for the faithfulness direction ($R-O$, the response to the pedestrian) and $\hat{v}_I$ for the interference direction ($N-O$, the response to a change carrying no driving-relevant information); these are unit vectors in representation space, the activation-side counterparts of the

behavioural F and I of Section VII. The two are not merely uncorrelated: in aggregate they oppose one another (mean cosine -0.54 over four policies), so a policy encodes “it got dark” as it encodes “the hazard is gone”. We had expected policies with more orthogonal directions to shift less between domains; they shifted more (Table VI, P6), and alignment itself changes with the driving side for all but AutoVLA. The magnitude of the lighting effect is a trait a small check-up can measure; its direction is not, until hundreds of frames are available (Section VI).

IX. Discussion

a) Using the report: Read one exam at a time, the report answers separate questions that a single score merges. Whether a policy is safe because it yields or because it stays short is answered by exposure together with hazard sensitivity: DiffusionDrive keeps its distance by planning short trajectories, which holds only as long as the trajectories stay short, whereas AutoVLA and Alpamayo-1.5 cover a human-like distance without the response to match. Whether a plan that moves is a plan that yields is answered by the decomposition of Section VII: among the cells in which the pedestrian-blind plan does reach the pedestrian, only 2.5% meet all three conditions, and displacement alone would have ranked the least safe policy first. The exam that no current benchmark reports is scaling: whether the response grows when the hazard does. Five policies fail it, and the one that passes, DiffusionDriveV2, does so at a sensitivity below 0.15.

b) Limitations: The exams are open-loop. The approach sequences are replayed from logs and re-anchored to the logged pose. The night is a synthetic rendering that also halves the number of other pedestrians the detector finds. The counterfactual pairs are constructed by editing, not captured, and so carry noise: 8.5% of removals leave a detectable residue, the inpainting artefact is 5–17% of the effect, and the VLA models can be edited only through their front camera. Alpamayo-1.5 covers 170 of the 236 frames. SimLingo plans 2.5s; for NAVSIM’s 4s we extrapolate at constant velocity (holding position instead gives EPDMS 0.47, still the lowest). Cross-policy correlations rest on six policies.

c) Future work: The check-up is designed as the first half of a benchmark-to-deployment pairing, and as one hazard type among several a deployment decision must cover: extending its exams to lead-vehicle braking and turning or intersection conflicts is the natural next scenario. Within the pedestrian scenario, the next step is to collect paired data rather than construct it: the same scene with and without a pedestrian, by day and by night. We then plan to validate the diagnoses against the same policies’ behaviour on a real vehicle. The sample-size laws above tell us how large that collection must be. A few dozen pairs suffice for planned distance, over-reaction and

lighting; hundreds of pairs with a genuine need to react are needed for hazard sensitivity.

X. Conclusion

A benchmark score reports an outcome, not a cause. We read the behaviour that a deployment decision needs from a few hundred logged frames, using counterfactual exams that an independent detector verifies. Three results are worth carrying forward. A near-zero counterfactual response is a reliable certificate that the pedestrian did not enter the decision, while a large response certifies nothing: of the plans that moved, as many moved towards the pedestrian as away from it. What transfers between driving sides is how steadily a policy drives rather than how strongly it reacts, and that exam is the only one—among ours and the standard scores alike—that predicts behaviour on the new side at significance. And each verdict can be priced in advance: some are settled in a dozen frames, others need hundreds, which is what a small check-up must know about itself before it is used to decide anything.

References

- [1] K. Chitta, A. Prakash, B. Jaeger, Z. Yu, K. Renz, and A. Geiger, “TransFuser: Imitation with transformer-based sensor fusion for autonomous driving,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2023.
- [2] B. Liao, S. Chen, H. Yin, B. Jiang, C. Wang, S. Yan, X. Zhang, X. Li, Y. Zhang, Q. Zhang, and X. Wang, “DiffusionDrive: Truncated diffusion model for end-to-end autonomous driving,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.
- [3] J. Zou, S. Chen, B. Liao, Z. Zheng, Y. Song, L. Zhang, Q. Zhang, W. Liu, and X. Wang, “DiffusionDriveV2: Reinforcement learning-constrained truncated diffusion modeling in end-to-end autonomous driving,” *arXiv preprint arXiv:2512.07745*, 2025.
- [4] K. Renz, L. Chen, E. Arani, and O. Sinavski, “SimLingo: Vision-only closed-loop autonomous driving with language-action alignment,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.
- [5] Z. Zhou, T. Cai, S. Z. Zhao, Y. Zhang, Z. Huang, B. Zhou, and J. Ma, “AutoVLA: A vision-language-action model for end-to-end autonomous driving with adaptive reasoning and reinforcement fine-tuning,” *arXiv preprint arXiv:2506.13757*, 2025.
- [6] NVIDIA, “Alpamayo-R1: Bridging reasoning and action prediction for generalizable autonomous driving in the long tail,” *arXiv preprint arXiv:2511.00088*, 2025.
- [7] —, “Alpamayo 1.5: An open reasoning vision-language-action model with navigation guidance,” 2026, model card and code, <https://github.com/NVlabs/alpamayo1.5>.
- [8] D. Dauner, M. Hallgarten, T. Li, X. Weng, Z. Huang, Z. Yang, H. Li, I. Gilitschenski, B. Ivanovic, M. Pavone, A. Geiger, and K. Chitta, “NAVSIM: Data-driven non-reactive autonomous vehicle simulation and benchmarking,” in *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2024.
- [9] W. Cao, M. Hallgarten, T. Li, D. Dauner, X. Gu, C. Wang, Y. Miron, M. Aiello, H. Li, I. Gilitschenski, B. Ivanovic, M. Pavone, A. Geiger, and K. Chitta, “Pseudo-simulation for autonomous driving,” in *Conference on Robot Learning (CoRL)*, 2025.
- [10] Z. Li, Z. Yu, S. Lan, J. Li, J. Kautz, T. Lu, and J. M. Alvarez, “Is ego status all you need for open-loop end-to-end autonomous driving?” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.

- [11] M. T. Ribeiro, T. Wu, C. Guestrin, and S. Singh, “Beyond accuracy: Behavioral testing of NLP models with CheckList,” in *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2020.
- [12] D. Hendrycks and T. Dietterich, “Benchmarking neural network robustness to common corruptions and perturbations,” in *International Conference on Learning Representations (ICLR)*, 2019.
- [13] L. Kong, Y. Liu, X. Li, R. Chen, W. Zhang, J. Ren, L. Pan, K. Chen, and Z. Liu, “Robo3D: Towards robust and reliable 3D perception against corruptions,” in *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.
- [14] S. Xie, L. Kong, W. Zhang, J. Ren, L. Pan, K. Chen, and Z. Liu, “RoboBEV: Towards robust bird’s eye view perception under corruptions,” *arXiv preprint arXiv:2304.06719*, 2023.
- [15] P. de Haan, D. Jayaraman, and S. Levine, “Causal confusion in imitation learning,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.
- [16] C. Wen, J. Lin, T. Darrell, D. Jayaraman, and Y. Gao, “Fighting copycat agents in behavioral cloning from observation histories,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- [17] F. Codevilla, E. Santana, A. M. López, and A. Gaidon, “Exploring the limitations of behavior cloning for autonomous driving,” in *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019.
- [18] R. Geirhos, J.-H. Jacobsen, C. Michaelis, R. Zemel, W. Brendel, M. Bethge, and F. A. Wichmann, “Shortcut learning in deep neural networks,” *Nature Machine Intelligence*, vol. 2, pp. 665–673, 2020.
- [19] P. Jacob, É. Zablocki, H. Ben-Younes, M. Chen, P. Pérez, and M. Cord, “STEEX: Steering counterfactual explanations with semantics,” in *European Conference on Computer Vision (ECCV)*, 2022.
- [20] M. Zemni, M. Chen, É. Zablocki, H. Ben-Younes, P. Pérez, and M. Cord, “OCTET: Object-aware counterfactual explanations,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023.
- [21] A. Zou, L. Phan, S. Chen, J. Campbell, P. Guo, R. Ren, A. Pan, X. Yin, M. Mazeika, A.-K. Dombrowski et al., “Representation engineering: A top-down approach to AI transparency,” *arXiv preprint arXiv:2310.01405*, 2023.
- [22] A. Azaria and T. Mitchell, “The internal state of an LLM knows when it’s lying,” in *Findings of the Association for Computational Linguistics: EMNLP*, 2023.
- [23] A. Kirillov, E. Mintun, N. Ravi, H. Mao, C. Rolland, L. Gustafson, T. Xiao, S. Whitehead, A. C. Berg, W.-Y. Lo, P. Dollár, and R. Girshick, “Segment anything,” in *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.
- [24] R. Suvorov, E. Logacheva, A. Mashikhin, A. Remizova, A. Ashukha, A. Silvestrov, N. Kong, H. Goka, K. Park, and V. Lempitsky, “Resolution-robust large mask inpainting with fourier convolutions,” in *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2022.
- [25] S. Ren, K. He, R. Girshick, and J. Sun, “Faster R-CNN: Towards real-time object detection with region proposal networks,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2015.
- [26] E. T. Hall, *The Hidden Dimension*. Doubleday, 1966.
- [27] D. Gettman and L. Head, “Surrogate safety measures from traffic simulation models,” *Transportation Research Record*, vol. 1840, no. 1, pp. 104–115, 2003.
- [28] J. C. Hayward, “Near-miss determination through use of a scale of danger,” in *Highway Research Record*, no. 384, 1972, pp. 24–34.
- [29] T. Kruse, A. K. Pandey, R. Alami, and A. Kirsch, “Human-aware robot navigation: A survey,” *Robotics and Autonomous Systems*, vol. 61, no. 12, pp. 1726–1743, 2013.
- [30] P. Hase, M. Bansal, B. Kim, and A. Ghandeharioun, “Does localization inform editing? surprising differences in causality-based localization vs. knowledge editing in language models,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [31] H. Caesar, V. Bankiti, A. H. Lang, S. Vora, V. E. Liong, Q. Xu, A. Krishnan, Y. Pan, G. Baldan, and O. Beijbom, “nuScenes: A multimodal dataset for autonomous driving,” in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020.